THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 4: **Changes Around Us: Physical and Chemical** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. During the burning of a candle, heat and light are given out. This suggests: 1

(a) no change **(b)** a reversible change
(c) a chemical change **(d)** a physical change

2. Why does the silver spoon get black while stainless steel spoons remain bright? 1

(a) steel is a gas **(b)** silver is a non-metal
(c) silver reacts with sulphur in air **(d)** steel does not

3. Which observation would show a chemical change in a mixture of sand and iron filings on heating with sulphur? 1

(a) iron attracted by magnet
(b) colour same
(c) a new black solid not attracted by a magnet
(d) sand settles

4. Which of these changes absorbs energy? 1

(a) melting of ice **(b)** rusting **(c)** burning of wood **(d)** respiration

5. Why are water pipes made of galvanised iron or plastic? 1

(a) to be soft **(b)** to resist rusting
(c) to be heavy **(d)** to be expensive

6. Which of the following correctly explains why a rusted nail breaks easily? 1

(a) rust is liquid **(b)** rust is magnetic
(c) rust is shiny **(d)** rust is brittle and flakes off

7. Which of these changes releases energy? 1

(a) evaporation of water **(b)** dissolving sugar
(c) melting of wax **(d)** burning of LPG

8. Why is cooking gas LPG burning considered a chemical change? 1

(a) it freezes **(b)** it evaporates
(c) new products carbon dioxide and water **(d)** it melts

9. **Assertion (A):** Burning of a candle involves both a physical and a chemical change. 1
Reason (R): The wax melts (physical) and the wax vapour burns to form new substances (chemical).
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Dissolving sugar in water is a chemical change. 1
Reason (R): Sugar can be recovered from the solution by evaporating the water.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is it necessary to use boiled water in the rusting experiment? 2

12. Why is the cooking of vegetables in a pressure cooker faster? 2

13. Why is the colour change on heating copper sulphate crystals reversible while burning of paper is not? 2

14. Suggest how to test that gas from baking soda and vinegar is not oxygen. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how ripening of a fruit can be speeded up or slowed and why it is chemical. 3

16. How are rusting and burning similar and different? 3

17. Why do leaves change colour in autumn? 3

OR

When ice melts, its temperature stays at 0 °C till all ice melts. Explain whether the process is physical or chemical. 3

18. Explain why vinegar can be used to clean a kettle and name the type of change. 3

OR

Explain the term 'irreversible' using two examples and why they cannot be reversed. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** The iron gate of Rajesh's house in coastal Mumbai has developed brown, flaky patches. Rajesh finds that the paint at those places has peeled off. His neighbour's gate, which was coated with zinc, looks fine. 5
(i) Name the brown substance on the gate.
(ii) What two things are needed for its formation?
(iii) Is the change physical or chemical?
(iv) Name two ways to protect the gate and explain how they work.

20. Write a note on rusting and its economic impact; propose five national-level steps to reduce the loss. 5

OR

Explain the changes in the activity of making a clay pot from wet clay, drying, and firing. 5

END OF QUESTION PAPER — Total: 40 Marks